

Deep Learning for Natural Language Processing

Assignment- 0 Solution

TYPE OF QUESTION: MCQ (Single Correct)

Number of questions: 10

Total marks: $10 * 1 = 10$

1. In NLP, we often represent words as normalized vectors. If the dot product of two normalized word vectors u and v is 0, what does this imply about their relationship in the vector space?

- A) They are identical (synonyms).
- B) They point in opposite directions (antonyms).
- C) They are orthogonal (uncorrelated or independent).
- D) One vector is a scaled version of the other.

Answer - C

2. Consider a single layer in a neural network where the input x is a row vector of shape $(1, d)$ and the weight matrix W has shape (d, k) . What is the primary geometric effect of the operation xW ?

- A) It calculates the dot product between the input and every column of W , projecting x into a k -dimensional feature space.
- B) It rotates the input vector x within the original d -dimensional space without changing its dimensionality.
- C) It computes the cross product, resulting in a vector orthogonal to both x and the columns of W .
- D) It scales the input vector x by the determinant of W .

Answer - A

3. When applying Principal Component Analysis (PCA) to reduce the dimensionality of word embeddings, we examine the eigenvalues of the covariance matrix. What does a very small (near-zero) eigenvalue indicate?

- A) The corresponding dimension contains the maximum amount of variance (information).

- B) The data is completely uncorrelated in that dimension.
- C) The corresponding eigenvector direction contains very little variance (noise or redundancy) and can likely be discarded.
- D) The matrix is non-invertible and the data cannot be processed.

Answer - C

4. Why is it often desirable to initialize the weight matrices of Recurrent Neural Networks (RNNs) to be orthogonal matrices?

- A) Orthogonal matrices ensure all outputs are positive, which fits the ReLU activation function.
- B) Orthogonal transformations preserve the vector norm (length), helping to prevent vanishing or exploding gradients over many time steps.
- C) Orthogonal matrices force the word vectors to be sparse (mostly zeros).
- D) It ensures that the dot product of the weights is always 1.

Answer - B

5. If your learned word embedding matrix (vocabulary size $\times$ dimension) has a rank significantly lower than the embedding dimension, what does this suggest?

- A) The embedding dimension is too small to capture the vocabulary's complexity.
- B) The dimensions are linearly dependent, meaning some features are redundant and the effective dimensionality is lower than configured.
- C) The model has perfectly memorized the training data.
- D) The matrix is sparse and computationally efficient.

Answer - B

6. Why is Cosine Similarity often preferred over Euclidean Distance for measuring the similarity between document vectors in NLP?

- A) Cosine similarity is computationally faster to calculate.
- B) Cosine similarity focuses on the orientation (topic) of the vectors rather than their magnitude (document length).
- C) Euclidean distance cannot handle negative numbers.

D) Cosine similarity always results in a positive number.

Answer - B

7. Mathematically, the gradient of a loss function, $\nabla L(\theta)$, points in the direction of steepest ascent. Why, then, is the update rule in Gradient Descent $\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla L(\theta)$?

A) Because we want to maximize the loss function to improve accuracy.

B) Because the gradient points to the global minimum, so we subtract it to get there faster.

C) Because we want to minimize the loss, and moving in the direction opposite to the gradient is the direction of steepest descent.

D) Because the learning rate α is usually negative, so the subtraction effectively becomes addition.

Answer - C

8. Which calculus concept is the immediate mathematical foundation for the Backpropagation algorithm used to train neural networks?

A) Integration by parts.

B) The Chain Rule, which allows us to compute the derivative of a composite function by multiplying the derivatives of its nested parts.

C) The Mean Value Theorem.

D) Taylor Series expansion for approximation.

Answer - B

9. In a neural network, calculating the partial derivative of the Loss L with respect to a specific weight w_{ij} , denoted as $\partial L / \partial w_{ij}$, tells us specifically:

A) The total error of the network at that specific neuron.

B) How much the Loss L would change if we made a tiny change to w_{ij} , assuming all other weights are held constant.

C) The value w_{ij} should take to make the Loss zero immediately.

D) The average rate of change of the Loss over the entire training epoch.

Answer - B

10. The Hessian matrix contains second-order partial derivatives. If the Hessian of the loss function is positive definite at a critical point (where the gradient is zero), what kind of point is it?

- A) A saddle point (minima in some directions, maxima in others).
- B) A global maximum.
- C) A local minimum (the surface curves upward in all directions).
- D) An inflection point.

Answer - C